Towards **Reliable Machine Learning** for **Visual Analytics**

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I tackle **prevalent assumptions** that makes
ML applications in visual analyatics unreliable.

I **design** and **democratize** diverse techniques for the purpose

I focus on **dimensionality reduction** and **clustering**,
two ML algorithms that are widely used for visual analytics

Reliable Algorithms

- UMATO (**VIS 2022**)

Reliable Interactions

- CLAMS (**VIS 2023**)
  Best papers honorable mention (top 5%)
- Distortion-aware brushing (**wip**)
  under review (**TVCG**)

Reliable Evaluations

- Steadiness & Cohesiveness (**VIS 2021**)
- Classes are not Clusters (**VIS 2023**)
- Adjusted IVMs (**wip**)
  under review (**TPAMI**)

Democratizing the Findings

- ZADU (**VIS 2023**)
- Structural Complexity (**wip**)
  under review (**VIS 2024**)
- Why not be reliable? (**wip**)

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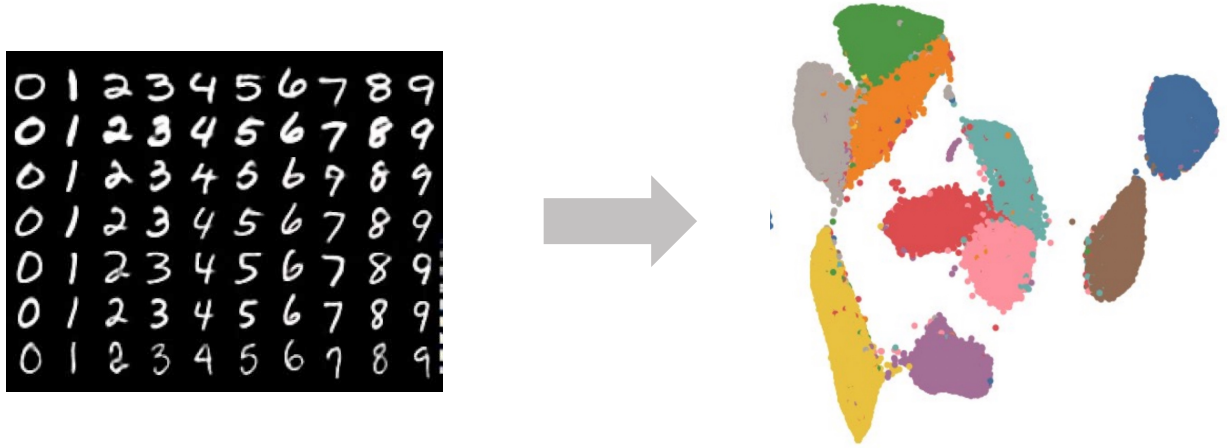


Reliable Algorithms

UMATO IEEE VIS 2022

UMAP is widely used for **cluster analysis**

However, the technique actually focus on **neighborhood structure**, thus using it for cluster analysis is unreliable



UMATO IEEE VIS 2022

I designed **UMATO,** a dimensionality reduction technique that reliably depicts both **neighborhood** and **cluster** structures



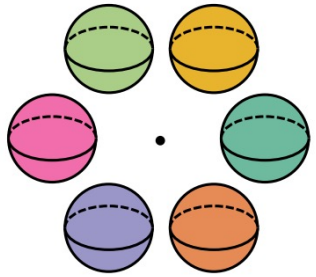




Reliable Evaluations

Class Labels are widely used as well-separated **ground truth cluster** in evaluating **dimensionality reduction** and **clustering**

However, **class labels** often do not form **clear clusters**, making evaluations using the labels **less reliable**



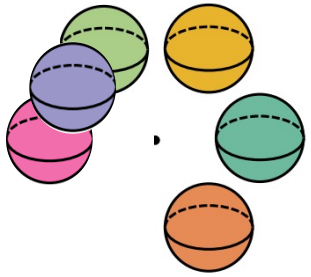
Class labels



Good embedding

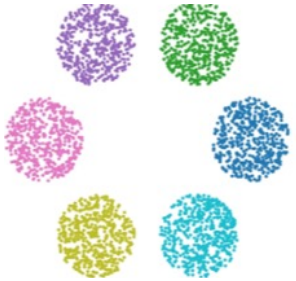


Bad embedding



Class labels

!= Ground Truth Clusters



Good embedding??

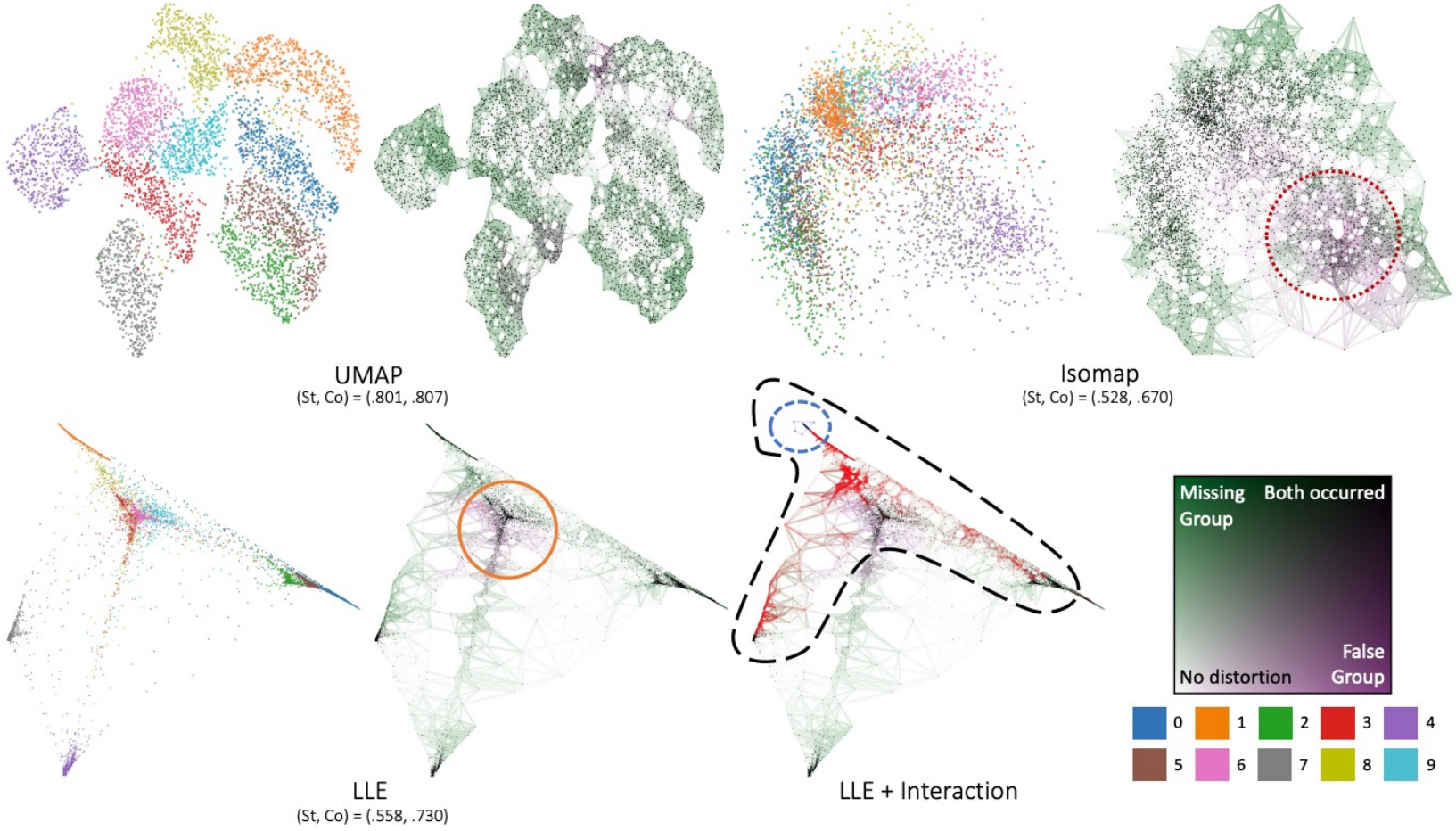






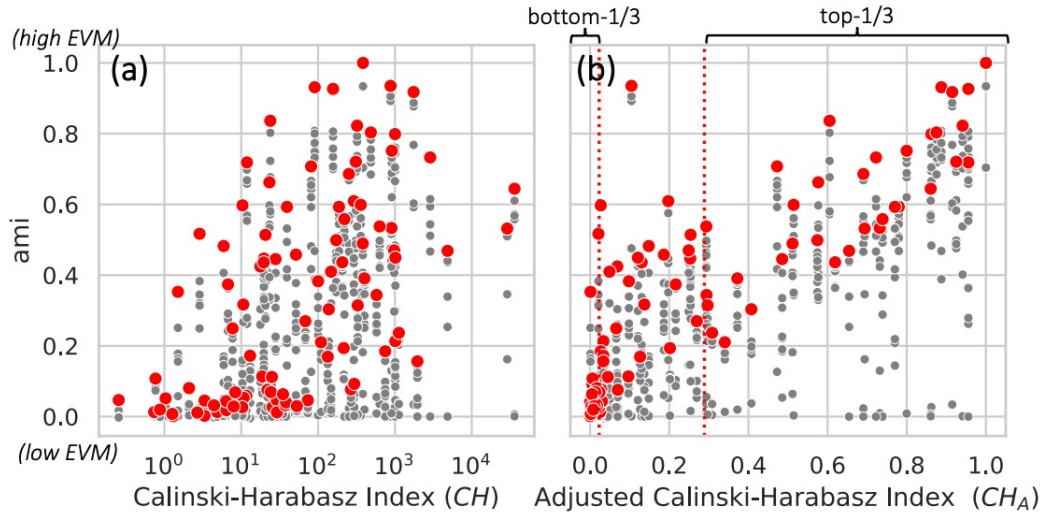
Bad embedding??

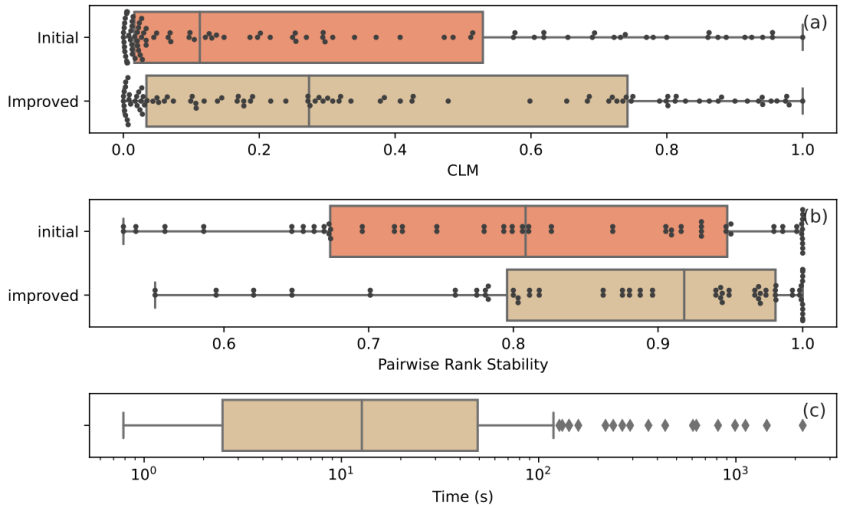
Steadiness & Cohesiveness IEEE VIS 2021



Steadiness & Cohesiveness IEEE VIS 2021

Adjusted IVMs *under review*, IEEE TPAMI

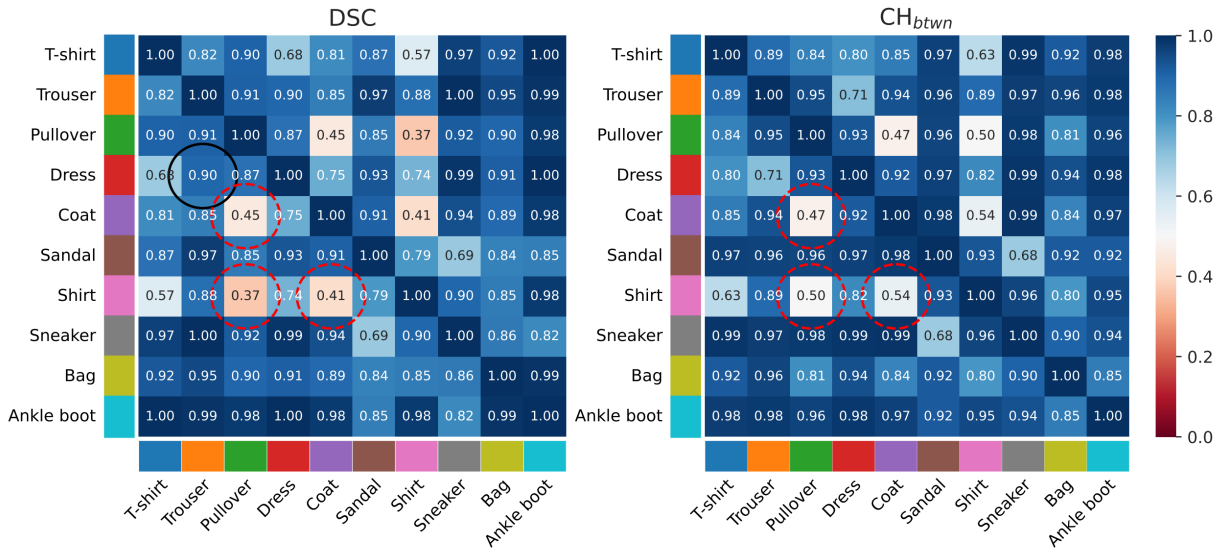




Steadiness & Cohesiveness IEEE VIS 2021

Adjusted IVMs *under review*, IEEE TPAMI

Classes are not Clusters IEEE VIS 2023



Steadiness & Cohesiveness IEEE VIS 2021

Let's **avoid using class labels!!**
Instead, let's use **clustering algorithms** to examine cluster structure

Adjusted IVMs *under review*, IEEE TPAMI

Let's check the separability of class labels
and **only use well-separated classes** as ground truth clusters

Classes are not Clusters IEEE VIS 2023

Let's check the separability of class labels
and use the **separability itself as ground truth**

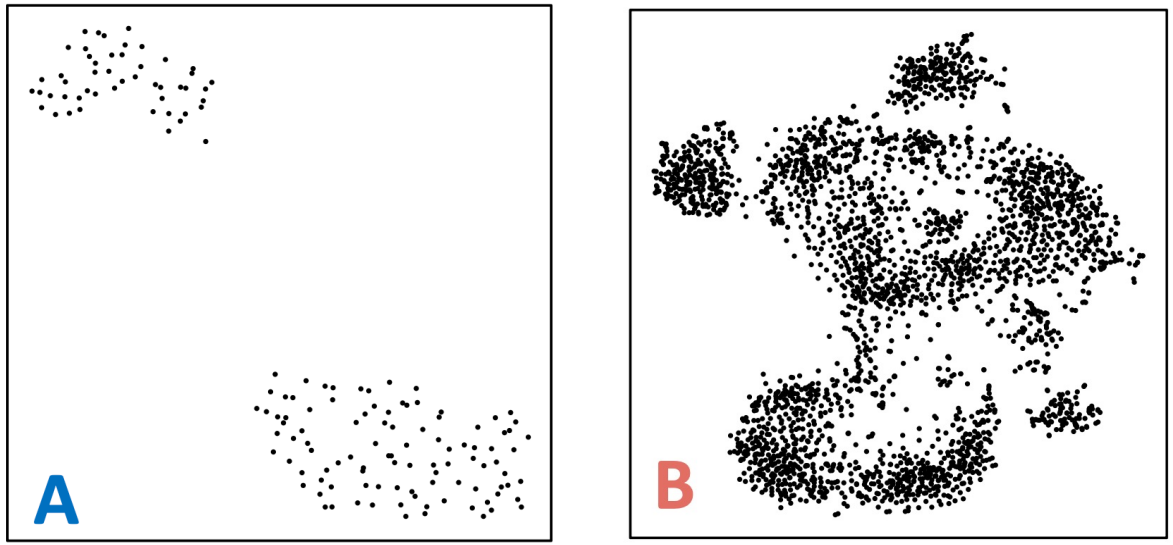


Reliable Interactions

CLAMS IEEE VIS 2023 *Best papers honorable mention (top 5%)*

Inconsistency between analysts can make visual analytics unreliable

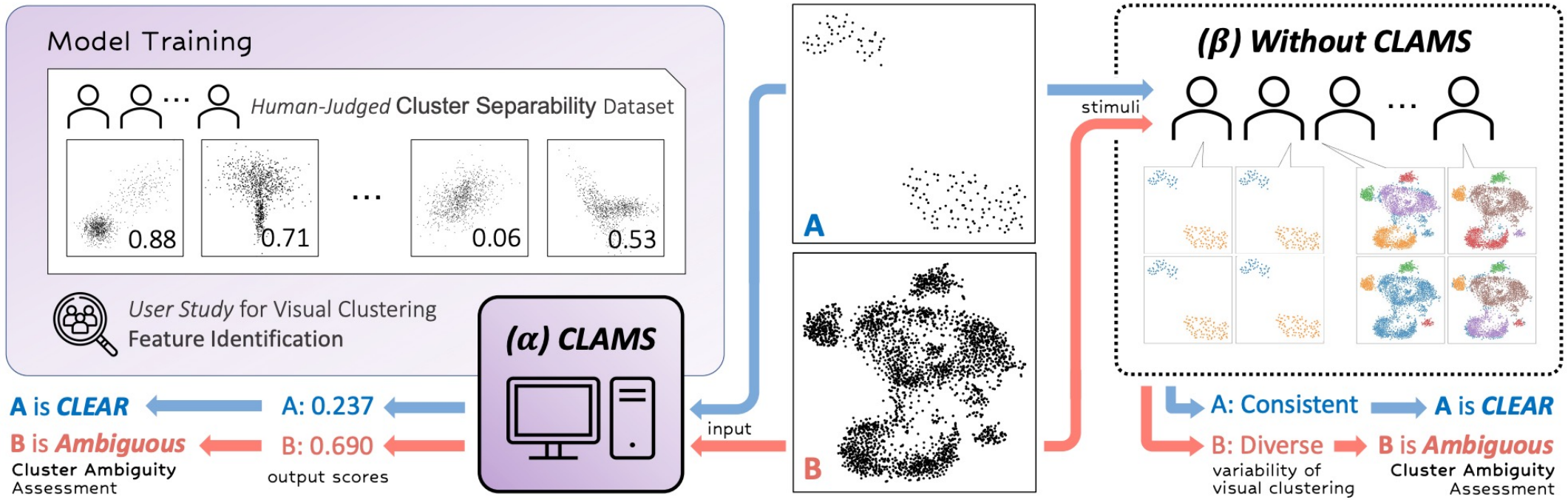
Ambiguous visualizations incurs such inconsistency



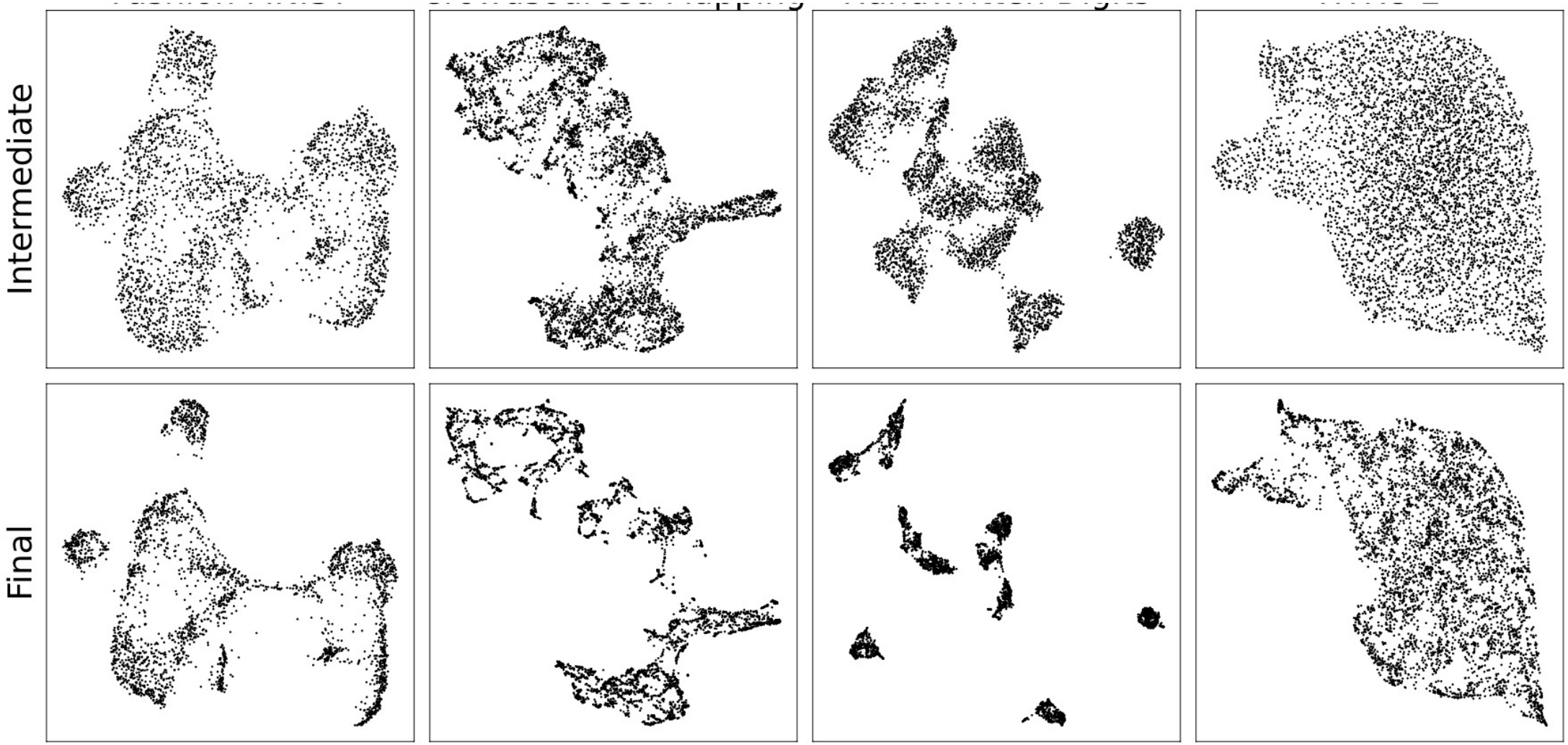
Clear Ambiguous

CLAMS IEEE VIS 2023

I designed **CLAMS,** a visual quality metrics that automatically quantifies the ambiguity of monochrome scatterplots



CLAMS + Dimensionality reduction





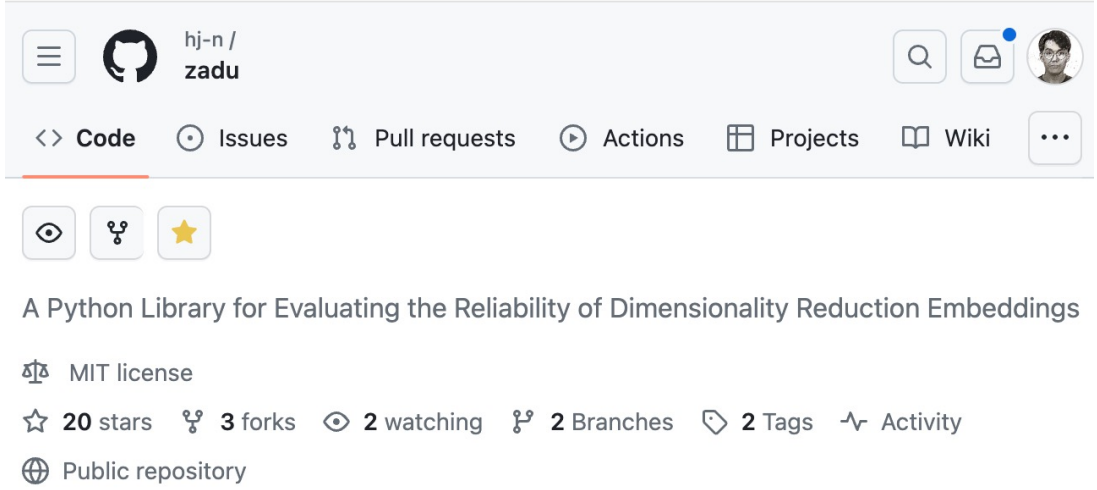
Democratizing the Findings

Making ML application reliable needs **additional effort**

How we can ***democratize our findings?***
How we can make **more practitioners to use ML more reliably?**

ZADU IEEE VIS 2023

Solution 1: Make techniques more accessible




```python
from zadu import zadu

hd, ld = load_datasets()
spec = [{
    "id"    : "tnc",
    "params": { "k": 20 },
}, {
    "id"    : "snc",
    "params": { "k": 30, "clustering_strategy": "dbscan" }
}]

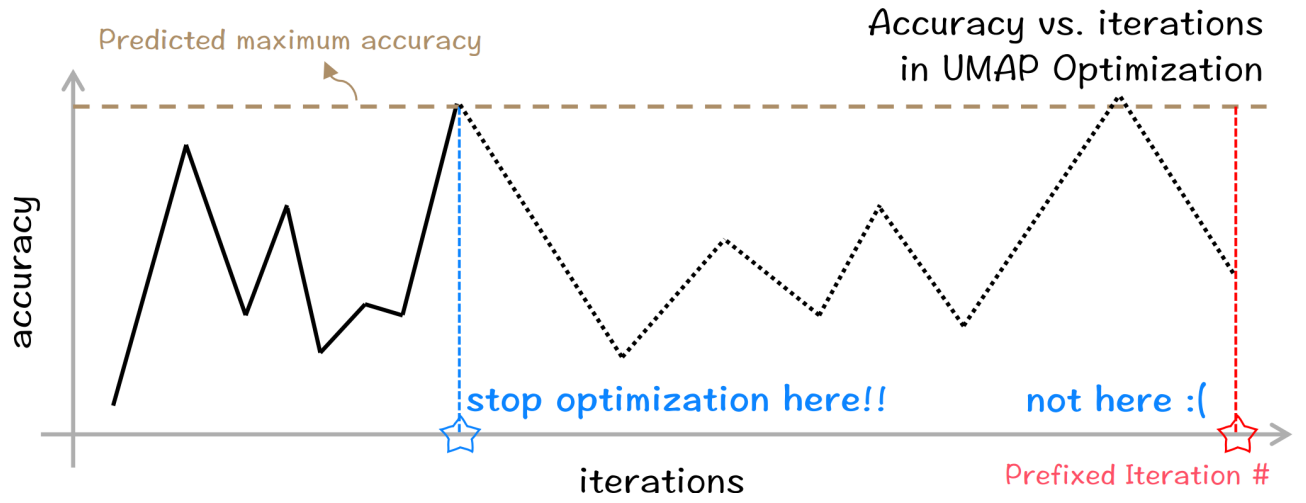
scores = zadu.ZADU(spec, hd).measure(ld)
print("T&C:", scores[0])
print("S&C:", scores[1])
```

ZADU IEEE VIS 2023

Solution 1: Make techniques more accessible

Structural Complexity *under review*, IEEE VIS 2024

Solution 2: Make techniques more efficient



ZADU IEEE VIS 2023

Solution 1: Make techniques more accessible

Structural Complexity *under review*, IEEE VIS 2024

Solution 2: Make techniques more efficient

Future work for dissertation:

Additional Solutions??

Future Projects (Maybe after graduation)

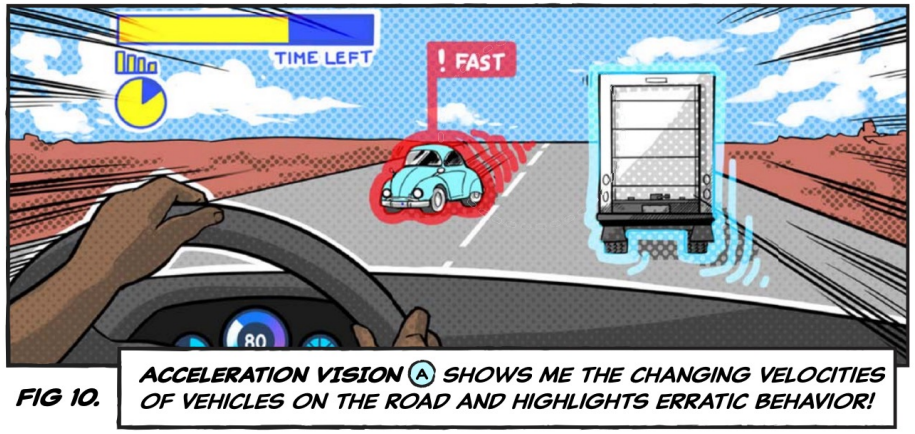
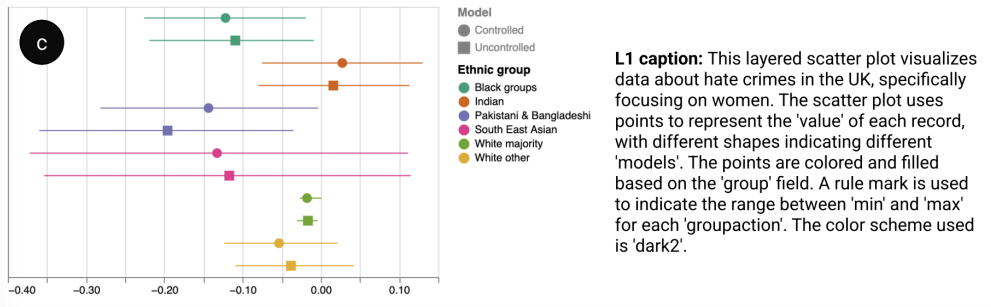


FIG 10.

Reliability problems / Evaluations
in **Immersive Analytics**



Reliability problems / Evaluations
in using **LLMs for visualization**

Willett, Wesley, et al. "Perception! immersion! empowerment! superpowers as inspiration for visualization." *IEEE transactions on visualization and computer graphics* 28.1 (2021): 22-32.

Acknowledgements

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